

4. Implement k-fold cross-validation on a classification model.

CODE A:

```
import pandas as pd
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier

print("Libraries Loaded.")

# Load Dataset
df = pd.read_csv("IRIS DATASET.csv")
print("Dataset Loaded.")

# Display first 5 rows (optional)
print(df.head())

# Features and Target
# Assuming the last column is the target (Species)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

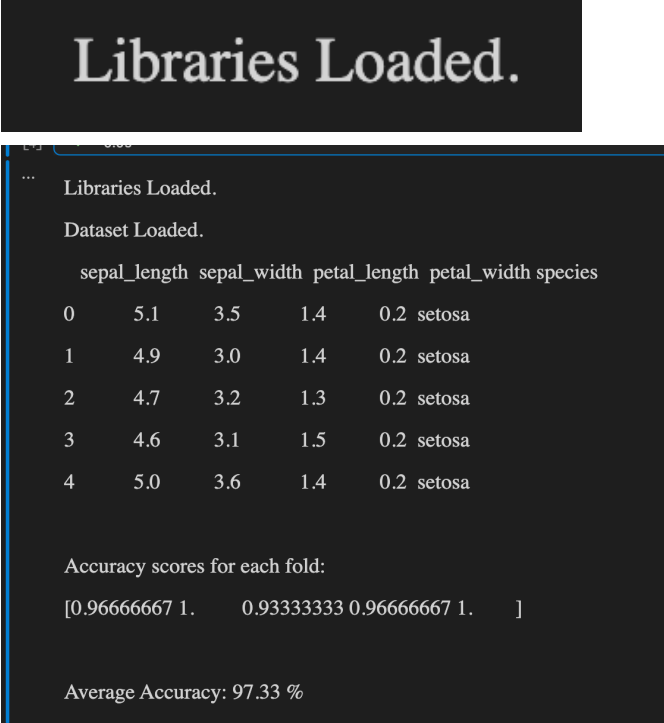
# Create KNN Classification Model
model = KNeighborsClassifier()

# Apply 5-Fold Cross Validation
scores = cross_val_score(model, X, y, cv=5)

# Print Accuracy Scores
print("\nAccuracy scores for each fold:")
print(scores)

# Print Average Accuracy
print("\nAverage Accuracy:", round(scores.mean() * 100, 2), "%")
```

OUTPUT:



```
Libraries Loaded.

Dataset Loaded.

  sepal_length sepal_width petal_length petal_width species
0         5.1         3.5         1.4         0.2  setosa
1         4.9         3.0         1.4         0.2  setosa
2         4.7         3.2         1.3         0.2  setosa
3         4.6         3.1         1.5         0.2  setosa
4         5.0         3.6         1.4         0.2  setosa

Accuracy scores for each fold:
[0.96666667 1. 0.93333333 0.96666667 1. ]

Average Accuracy: 97.33 %
```

CODE B:

```
#7.4.B
import pandas as pd
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsRegressor

print("Libraries Loaded.")

# Load Dataset
df = pd.read_csv("housing.csv")
print("Dataset Loaded.")

# Remove rows with missing values
df = df.dropna()

# Convert categorical column into numeric values
df = pd.get_dummies(df, columns=["ocean_proximity"], drop_first=True)

# Features and Target
X = df.drop("median_house_value", axis=1)
y = df["median_house_value"]

# Create KNN Regression Model
model = KNeighborsRegressor()

# Apply 5-Fold Cross Validation
scores = cross_val_score(model, X, y, cv=5, scoring="r2")

# Print Results
print("\nR2 Score for each fold:")
print(scores)

print("\nAverage R2 Score:", round(scores.mean(), 4))
```

OUTPUT:

```
... Libraries Loaded.

Dataset Loaded.

R2 Score for each fold:
[-0.00282638  0.25443877  0.19491464  0.03164222  0.21671463]

Average R2 Score: 0.139
```